

AURORA

AURORA: Attention Universes for Risk Observation and Review Allocation

News can fill a fifth to a quarter of a risk team's daily review list while keeping over 93% of the price-only list's tail-loss coverage, and every added name arrives with the source sentence that justifies it.

93.81%

tail-loss coverage kept

K = 10 review slots; 19.69% of names new

2,004,888

article–security–date records

6,431 securities · 1,699 trading days

94.32%

of records yield a complete evidence card

1,890,963 of 2,004,888, each citing its sentences

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Working paper, 2026. All figures are as printed in the submitted paper.

Ten review slots, thousands of headlines

A one-minute read for someone new to the topic. Slides 03–09 give the evidence.

In plain terms

The problem Before a risk report or stress test, a risk team can source-check only a handful of names, say 10, while the news mentions thousands. A price-based screen can only rank the names it already tracks.

The idea Let price risk fill most of the slots (8 of 10) and let an LLM reading the news fill the rest. Every added name arrives with the exact sentences that justified it, so a reviewer can verify it in seconds.

The result Across 2019–2025 this swaps about a fifth to a quarter of the daily list while keeping 93.8% of the price-only list's tail-loss coverage; 94% of 2 million news records yielded a complete evidence card.

MY ROLE · WHERE THIS COULD BE USED

My role Second author.

Use case Risk-team triage, and any alert queue with more alerts than reviewers and a need for an audit trail.

Terms used in this deck

REVIEW CAPACITY K

The number of names a team can check per day; the paper uses $K = 10$ and $K = 20$.

TAIL-LOSS COVERAGE

The share of the largest subsequent losses that the review list would have flagged; higher is better.

EVIDENCE CARD

The LLM's structured output for one article: a category, a summary and the sentence IDs it relied on.

STRESS TEST / SCENARIO LAW

Re-pricing the book under simulated moves; a "law" is the assumed distribution (historical, Gaussian, Student-t).

CRPS

A score for a whole predicted distribution against what actually happened; lower is better.

ES95

Expected shortfall: the average loss in the worst 5% of cases.

Fixed review capacity against a news firehose

Before a risk report or stress test, a team can source-check only a handful of names. Which ones?

Why it matters

- Risk teams can review only K names before a stress test, while the news stream mentions thousands of securities.
- “A price screen can only rank the names it already ranks.” News points at other names, with sentences a reviewer can open.
- The LLM routes a name to review and explains why. It does not forecast returns, severity or weights.

WHAT THIS MEANS OUTSIDE THE PAPER

Fixed budget, ranked queue price risk and news compete for the same review slots.

Evidence with every pick each name arrives with the sentences that selected it.

Generalizes to any queue with more alerts than reviewers and a need for audit.

Data & setup

NEWS

Financial Modeling Prep stock news and press releases, 2019–2025

RECORDS

2,004,888 article–security units; 99.62% with headline and text

UNIVERSE

6,431 securities; 1,699 S&P 500 trading days, Apr 2019–Dec 2025

LLM

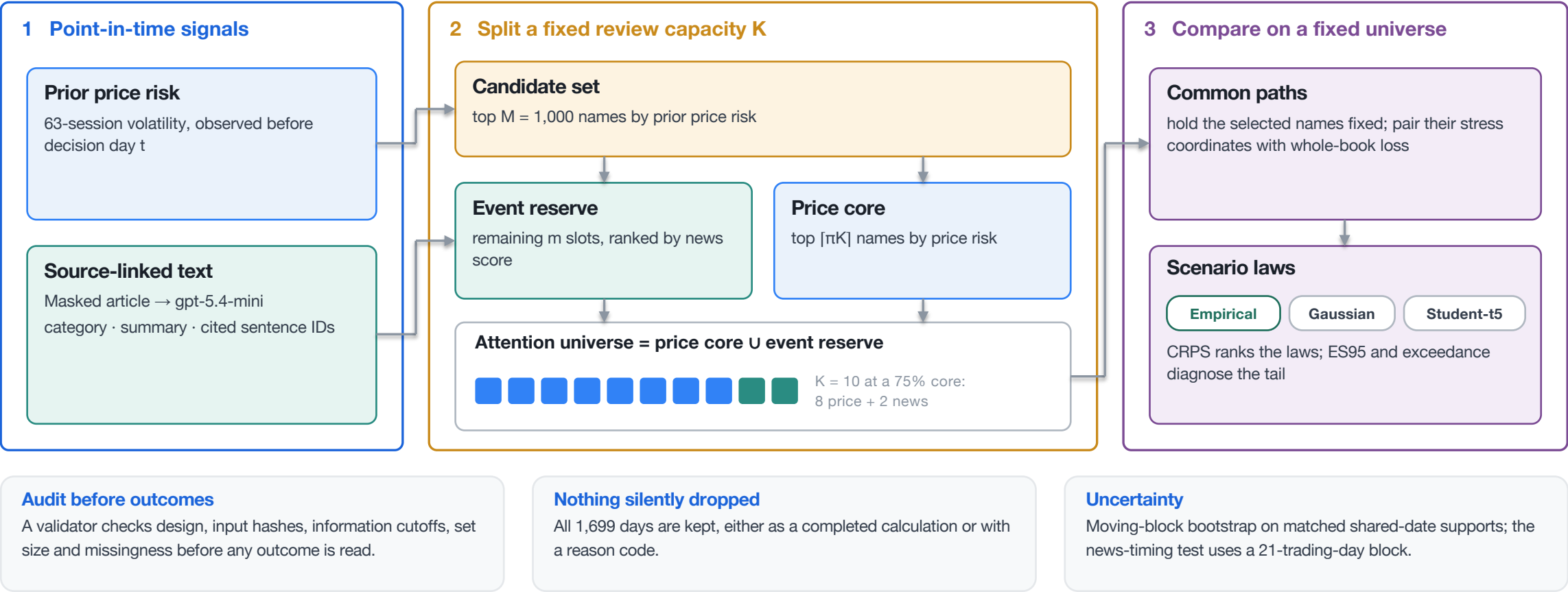
gpt-5.4-mini, prompt version article-issues-v3, hash-fixed

COVERAGE

1,890,963 complete structured records (94.32%)

How AURORA builds a daily review list

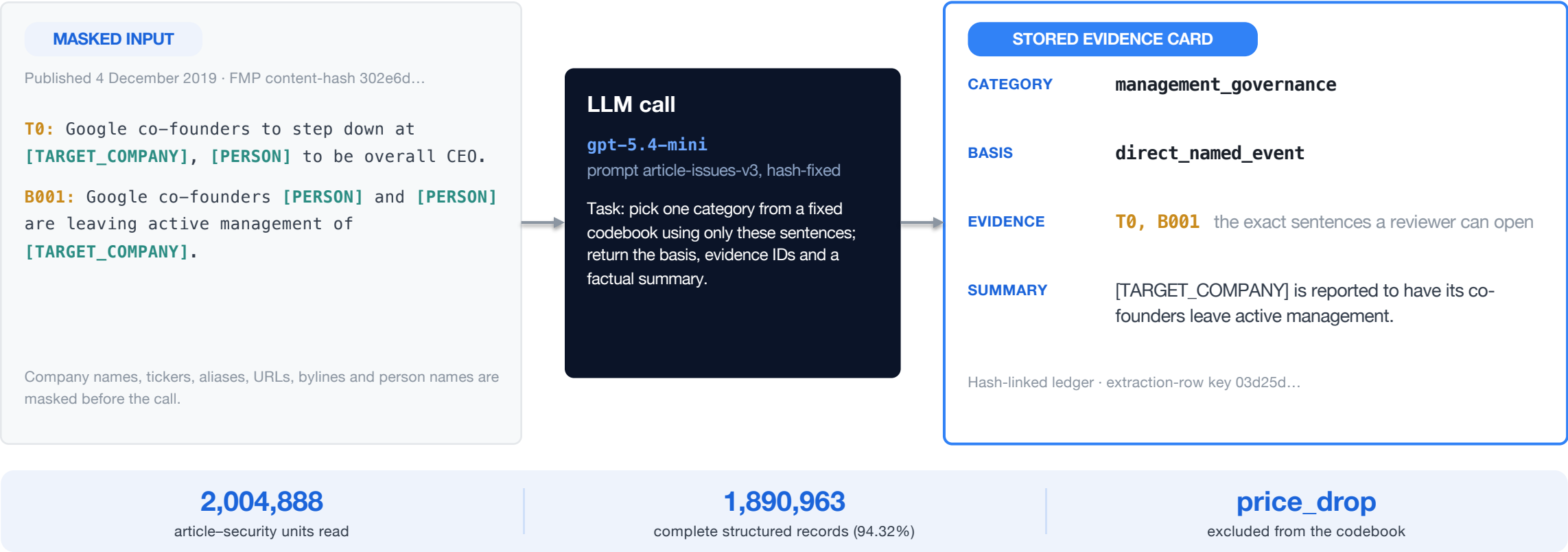
Price risk fills most review slots, LLM-structured news fills the rest, and the chosen names feed whole-book scenarios.



Redrawn and simplified from the paper's Figure 1.

Every news-flagged name comes with an evidence card

The LLM reads masked text and returns a category, the sentence IDs it relied on, and a summary a reviewer can check.

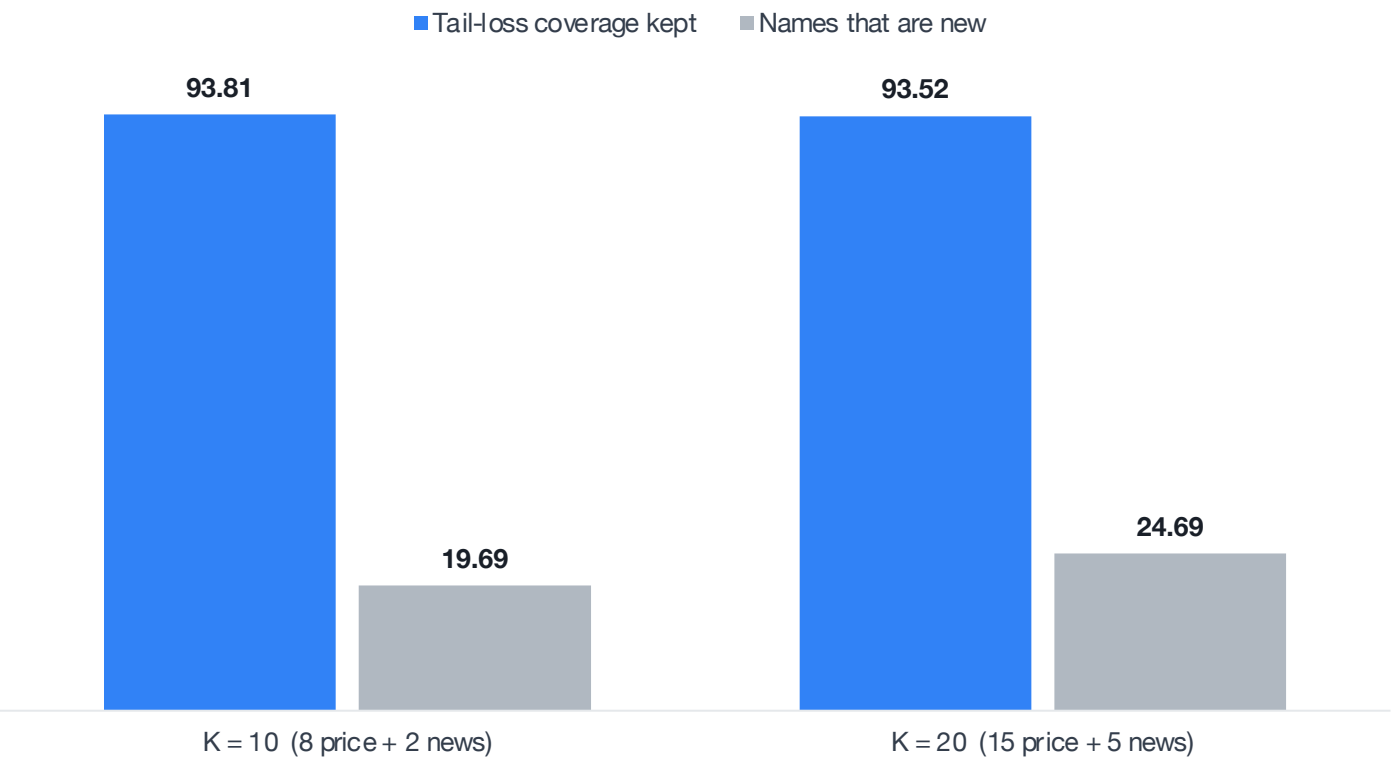


Example from the paper’s Table 1: an article published on 4 December 2019.

A quarter of the list changes; over 93% of tail coverage stays

At a 75% price core, news-ranked names fill the remaining review slots.

Same review capacity: coverage kept (higher is better) vs share of the list that changed



What it shows

- Coverage is measured against the price-only list at the same capacity, on 885 (K = 10) and 532 (K = 20) matched days.
- Every added news-ranked name has a complete evidence card with cited source sentences.

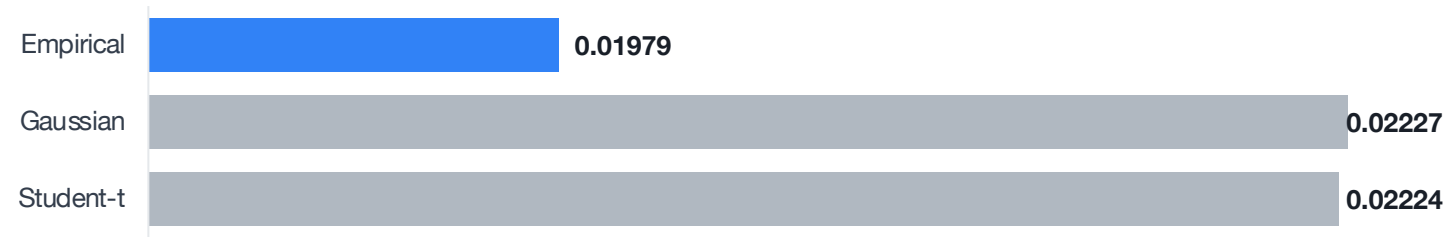
Source: AURORA Table 3 and Abstract.

Historical paths beat parametric scenario laws

Once the review list is chosen, the same names are stress-tested. Replaying real historical paths beats assuming a bell curve.

Scenario law	CRPS	Forecast ES95	Loss beyond ES95	Exceedance
Empirical (equal-weight)	0.01979	18.15%	16.92 bp	2.17%
Gaussian	0.02227	12.47%	33.16 bp	4.90%
Student-t	0.02224	13.31%	30.03 bp	4.31%

CRPS by scenario law (lower is better)



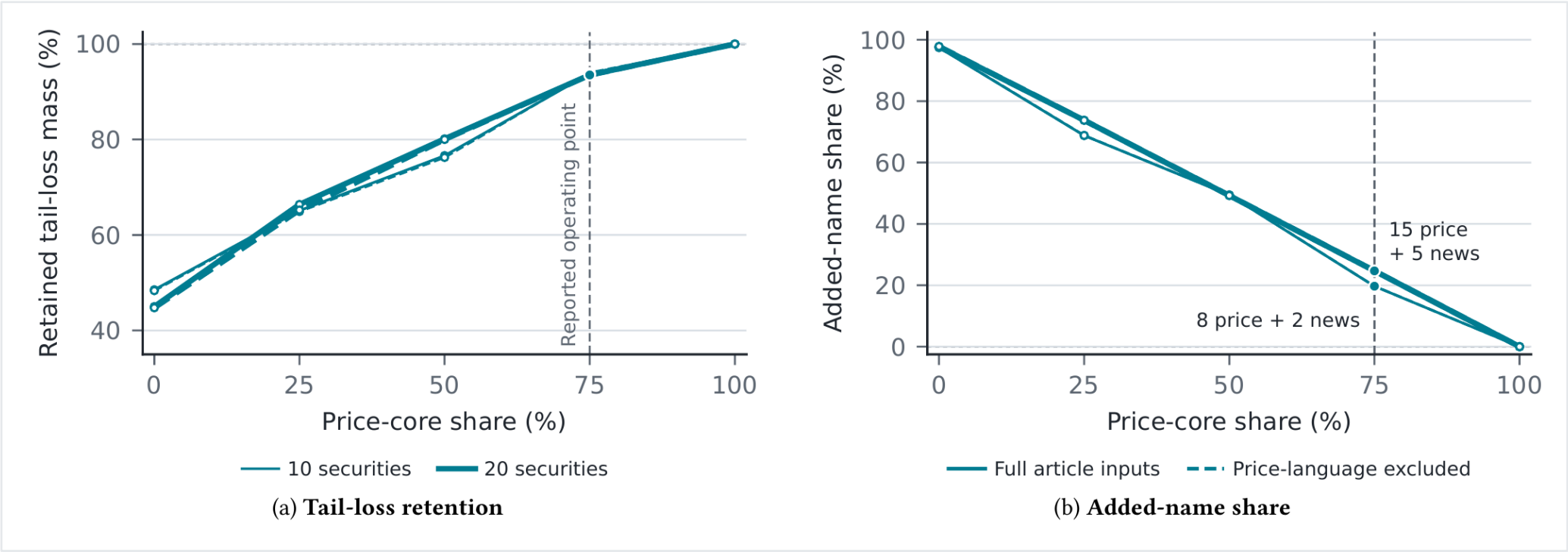
What it shows

- CRPS scores a whole predicted distribution against what actually happened; lower is better. It is the only number the laws are ranked on.
- CRPS falls 11.13% vs Gaussian and 11.01% vs Student-t(5).
- Laws are ranked by CRPS only; the expected-shortfall columns are diagnostics, not calibration tests.
- Test set: 1,677 common days and 83,631 date–security observations.

Source: AURORA Table 4 and Abstract.

The review frontier, as published

Original figures from the submitted paper, shown as published. Numbers on earlier slides are read from its tables and text.



Paper Figure 3: Price-core/news allocation frontier

Left: tail-loss coverage kept as the price-core share rises from 0% to 100%. Right: share of names that are new. The dashed line is the 75% operating point used on the results slides.

What held, and what the paper does not claim

The controls and the limits are both stated in the paper.

Checks that held

- Current vs 63-day-lagged news is statistically equivalent: median difference exactly 0, win share 0.5073, inside a band declared before outcomes were joined. The value of news is routing and evidence, not timing.
- Controls: lagged news, reverse rank, market cap and 20 time-shifted rankings.
- With price-reaction wording removed from the inputs, coverage is unchanged (93.99% / 93.50%).
- Entity masking and a hash-fixed prompt before every LLM call.
- The chosen names also cover the recorded events: 90.71% at $K = 10$ and 88.80% at $K = 20$.

What the paper does not claim

- News-covered names have a 2.44% ten-day tail rate vs 5.00% overall: news is “not a tail-risk alarm.”
- All results use the 2019–2025 development sample; a frozen future sample is still needed.
- The CRPS comparison was not repeated on a price-only universe, so it cannot be credited to news selection.
- The selected names are a review list, not a trading portfolio. No live trades were made.

Code (anonymous review repository): anonymous.4open.science/r/finance-risk-reproduction-34F7